# Monotonic Kolmogorov–Arnold Network (KAN) Transport Maps for L-SING: Theoretical Foundations and Monotonicity Proofs

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November 3, 2026

**Abstract**

In the Localized Sparsity Identification for Non-Gaussian Distributions (L-SING) framework [6], undirected graphical models are discovered by estimating local conditional transport maps $S^k(x_{-k}, \cdot)$ that push forward continuous conditional distributions $\pi(x_k \mid x_{-k})$ to a standard normal reference measure $\eta_k = \mathcal{N}(0,1)$. Existing implementations parameterize $S^k(x)$ using Unconstrained Monotonic Neural Networks (UMNNs) [14], which guarantee target monotonicity $\partial_k S^k > 0$ via numerical quadrature (e.g., 50-node Clenshaw–Curtis grids), imposing heavy computational overhead and requiring higher-order automatic differentiation through the numerical integrator. In this work, we rigorously examine the replacement of UMNNs with Monotonic Kolmogorov–Arnold Networks (KANs). We systematically identify the monotonicity requirements dictated by measure transport, proving that strict target monotonicity is fully enforceable across single-layer, two-layer, and arbitrarily deep ($L$-layer) KANs, while retaining unconstrained expressivity along conditioning dimensions $x_{-k}$. Furthermore, we prove that Scale-Modulated Monotonic KANs (SM-KANs) strictly preserve target monotonicity while overcoming the inherent additive polarization dilemma of standard KANs on multiplicative non-Gaussian dependencies. We present both an intuitive, accessible explanation and formal mathematical proofs, connecting our results with the recently established MonoKAN framework [11] and Fritsch–Carlson monotonicity conditions [4].

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# 1 The Measure Transport Formulation in L-SING

*Remark* 1.1 (Context and Scope). *Note for collaborator:* For completeness and clarity of notation, the following subsections provide a self-contained re-derivation of the L-SING conditional pullback density formula and the resulting learning loss function.

## 1.1 Conditional Transport Maps

Let $\eta_k = \mathcal{N}(0,1)$ denote the standard univariate Gaussian reference measure with Lebesgue density,

$$\eta_k(z_k) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} z_k^2\right). \tag{1}$$

For each node $k \in \{1, \dots, d\}$, L-SING seeks a conditional transport map component $S^k(x_{-k}, \cdot) : \mathbb{R} \to \mathbb{R}$, $x_k \mapsto z_k = S^k(x_{-k}, x_k)$, that pushes forward the conditional probability measure $\nu_{\pi(\cdot|x_{-k})}$ to the standard normal reference measure $\nu_{\eta_k}$,

$$S^k(x_{-k}, \cdot)_\sharp \, \nu_{\pi(\cdot|x_{-k})} = \nu_{\eta_k}. \tag{2}$$

By definition of the pushforward measure in set-integral form, for any Borel set $B \in \mathcal{B}(\mathbb{R})$,

$$\mathbb{P}\Big(Z_k \in B \;\Big|\; X_{-k} = x_{-k}\Big) = \mathbb{P}\Big(X_k \in (S^k(x_{-k}, \cdot))^{-1}(B) \;\Big|\; X_{-k} = x_{-k}\Big). \tag{3}$$

Expressed as Lebesgue integrals against their respective probability densities,

$$\int_B \eta_k(z_k)\, dz_k = \int_{(S^k(x_{-k},\cdot))^{-1}(B)} \pi(x_k \mid x_{-k})\, dx_k \quad \forall B \in \mathcal{B}(\mathbb{R}). \tag{4}$$

**[Key Requirement: 1. Target Monotonicity $\partial_k S^k > 0$]**
For the fiber map $x_k \mapsto S^k(x_{-k}, x_k)$ to be an orientation-preserving diffeomorphism on $\mathbb{R}$, its derivative along the target coordinate must be strictly positive everywhere,

$$\partial_k S^k(x) := \frac{\partial S^k(x)}{\partial x_k} > 0 \quad \forall x \in \mathbb{R}^d. \tag{5}$$

This ensures that the coordinate transformation $x_k \mapsto z_k$ is one-to-one, globally invertible on $\mathbb{R}$, and has a strictly positive Jacobian $|\partial_k S^k(x)| = \partial_k S^k(x) > 0$.

Under this strict monotonicity condition, we apply the 1D change-of-variables substitution $z_k = S^k(x_{-k}, x_k)$ to the right-hand side of (4). As $x_k$ traverses $(S^k(x_{-k}, \cdot))^{-1}(B)$, $z_k$ traverses $B$, with infinitesimal element $dz_k = \partial_k S^k(x_{-k}, x_k)\, dx_k$. Substituting $dx_k = \frac{1}{\partial_k S^k(x_{-k}, (S^k(x_{-k},\cdot))^{-1}(z_k))}\, dz_k$ yields,

$$\int_B \eta_k(z_k)\, dz_k = \int_B \frac{\pi\Big((S^k(x_{-k}, \cdot))^{-1}(z_k) \;\Big|\; x_{-k}\Big)}{\partial_k S^k\Big(x_{-k}, (S^k(x_{-k}, \cdot))^{-1}(z_k)\Big)}\, dz_k. \tag{6}$$

Rearranging into a single integral over $B$,

$$\int_B \left(\eta_k(z_k) - \frac{\pi\Big((S^k(x_{-k}, \cdot))^{-1}(z_k) \;\Big|\; x_{-k}\Big)}{\partial_k S^k\Big(x_{-k}, (S^k(x_{-k}, \cdot))^{-1}(z_k)\Big)}\right) dz_k = 0. \tag{7}$$

Since this identity must hold for **every** Borel set $B \in \mathcal{B}(\mathbb{R})$, the integrand vanishes Lebesgue-almost everywhere,

$$\eta_k(z_k) = \frac{\pi\Big((S^k(x_{-k},\cdot))^{-1}(z_k) \;\Big|\; x_{-k}\Big)}{\partial_k S^k\Big(x_{-k}, (S^k(x_{-k},\cdot))^{-1}(z_k)\Big)}. \tag{8}$$

Substituting $z_k = S^k(x)$ and solving for the conditional density yields the exact **conditional pullback density formula**,

$$\pi_{S^k}(x_k \mid x_{-k}) = \eta_k(S^k(x)) \cdot |\partial_k S^k(x)| = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(S^k(x))^2\right) \cdot \partial_k S^k(x). \tag{9}$$

Taking the natural logarithm yields the conditional log-density of the model,

$$\log \pi_{S^k}(x_k \mid x_{-k}) = -\frac{1}{2}(S^k(x))^2 + \log \partial_k S^k(x) - \frac{1}{2}\log(2\pi). \tag{10}$$

### 1.2 Connection to the Learning Objective and Functional Regularization

The training objective in L-SING is derived directly from minimizing the Kullback–Leibler (KL) divergence between the true data distribution $\pi$ and the model's pullback distribution $\pi_{S^k}$,

$$D_{\mathrm{KL}}(\pi \,\|\, \pi_{S^k}) = \mathbb{E}_{X\sim\pi}\left[\log \pi(X_k \mid X_{-k}) - \log \pi_{S^k}(X_k \mid X_{-k})\right] \tag{11}$$

$$= -\mathbb{E}_{X\sim\pi}\left[\log \pi_{S^k}(X_k \mid X_{-k})\right] + \text{const}, \tag{12}$$

where the constant depends only on the true data entropy and does not involve the map parameters. Substituting the conditional log-pullback formula (10) into this expected negative log-likelihood directly yields,

$$-\mathbb{E}_{X\sim\pi}\left[\log \pi_{S^k}(X_k \mid X_{-k})\right] = \mathbb{E}_{X\sim\pi}\left[\frac{1}{2}(S^k(X))^2 - \log \partial_k S^k(X)\right] + \frac{1}{2}\log(2\pi). \tag{13}$$

Dropping the irrelevant additive constant $\frac{1}{2}\log(2\pi)$ and replacing the expectation with an empirical Monte Carlo average over $M$ i.i.d. samples $\{x^{(i)}\}_{i=1}^M \sim \pi$, we obtain the empirical negative log-likelihood. To promote sparsity, a regularization term $\Phi(S^k)$ is added, leading to the regularized loss function,

$$\mathcal{L}(S^k) \coloneqq \underbrace{\frac{1}{M}\sum_{i=1}^{M}\left[\frac{1}{2}(S^k(x^{(i)}))^2 - \log \partial_k S^k(x^{(i)})\right]}_{\text{Empirical Negative Log-Pullback Likelihood } -\frac{1}{M}\sum_i \log \pi_{S^k}(x_k^{(i)}|x_{-k}^{(i)})} + \lambda\Phi(S^k), \tag{14}$$

which is minimized over the space of admissible maps, $\min_{S^k\in\mathcal{S}_k} \mathcal{L}(S^k)$, subject to the strict target monotonicity constraint $\partial_k S^k(x) > 0$ ($\pi$-a.e.). The regularization penalty $\Phi(S^k)$ is defined as,

$$\Phi(S^k) \coloneqq \sum_{j=1}^{d}\sqrt{\frac{1}{M}\sum_{i=1}^{M}\left(\frac{\partial S^k(x^{(i)})}{\partial x_j}\right)^2}. \tag{15}$$

### 1.3 The Generalized Precision Matrix

Spantini, Bigoni, and Marzouk [13] proved that for distributions with twice-differentiable log-densities, pairwise conditional independence is characterized by the vanishing of the mixed second derivatives of the log-density,

$$X_j \perp X_k \mid X_{-jk} \iff \partial_j\partial_k \log \pi(x) = 0 \quad \forall x \in \mathbb{R}^d. \tag{16}$$

Because $\partial_j \partial_k \log \pi(x) = \partial_j \partial_k \log \pi(x_k \mid x_{-k})$, the entries of the *generalized precision matrix* $\Omega \in \mathbb{R}^{d \times d}$ are defined as,

$$\Omega_{jk} \coloneqq \mathbb{E}\left|\partial_j \partial_k \log \pi(x_k \mid x_{-k})\right| = \mathbb{E}\left|\partial_j \partial_k \left[-\frac{1}{2}(S^k(x))^2 + \log \partial_k S^k(x)\right]\right|. \tag{17}$$

Using $N$ independent evaluation samples, the Monte Carlo estimator is,

$$\widehat{\Omega}_{jk} \coloneqq \frac{1}{N}\sum_{i=1}^{N}\left|\partial_j \partial_k \left[-\frac{1}{2}(S^k(x^{(i)}))^2 + \log \partial_k S^k(x^{(i)})\right]\right|. \tag{18}$$

The matrix is symmetrized via $\widehat{\Omega}_{\text{L-SING}} = \frac{1}{2}(\widehat{\Omega} + \widehat{\Omega}^T)$, the diagonal is fixed to $\widehat{\Omega}_{kk} = 1$, and values are normalized to a maximum of 1. Edges are determined by thresholding: $(j,k) \in E \iff |\widehat{\Omega}_{jk}| > \tau$.

**[Key Requirement: 2. Higher Regularity ($C^3$ Smoothness)]**

Evaluating the generalized precision entries $\Omega_{jk} = \mathbb{E}|\partial_j \partial_k \log \pi(x_k \mid x_{-k})|$ requires expanding the log-derivative integrand,

$$\begin{aligned}\partial_j \partial_k \left[-\frac{1}{2}(S^k(x))^2 + \log \partial_k S^k(x)\right] &= -(\partial_j S^k)(\partial_k S^k) - S^k(\partial_j \partial_k S^k) \\ &\quad + \frac{(\partial_j \partial_k^2 S^k)(\partial_k S^k) - (\partial_k^2 S^k)(\partial_j \partial_k S^k)}{(\partial_k S^k)^2}.\end{aligned} \tag{19}$$

Consequently, the transport map $S^k$ must possess continuous derivatives up to third order ($S^k \in C^3(\mathbb{R}^d)$, or piecewise $C^3$) so that the mixed third derivative $\partial_j \partial_k^2 S^k$ is well-defined.

**[Key Requirement: 3. Efficient Derivative Computability]**

Evaluating (19) across $N$ Monte Carlo evaluation samples and all $d^2$ pairs $(j,k)$ must be computationally tractable. Under numerical quadrature architectures (such as UMNNs), this requires backpropagating three times through a numerical integrator, creating massive memory footprints and runtime bottlenecks. The transport map should ideally admit closed-form, analytic derivatives to allow direct, instantaneous precision matrix extraction.

## 2 Limitations of Existing Architectures: MLPs and UMNNs

### 2.1 The Convexity Trap of Monotonic MLPs

In standard Multi-Layer Perceptrons (MLPs), enforcing monotonicity with respect to an input $x_k$ requires constraining network weights along that input's computational path to be non-negative ($W \geq 0$) and using non-decreasing activation functions (e.g., ReLU or Sigmoids). However, a well-known result by Archer and Wang [1] proves that constraining MLP weights to be non-negative with convex activation functions forces the entire neural network to be **globally convex** [8].

This creates a fundamental bottleneck for transport maps: a convex function cannot model sigmoidal transitions, S-curves, multi-modal density transformations, or higher-order inflections without introducing severe model bias (see Figure 1).

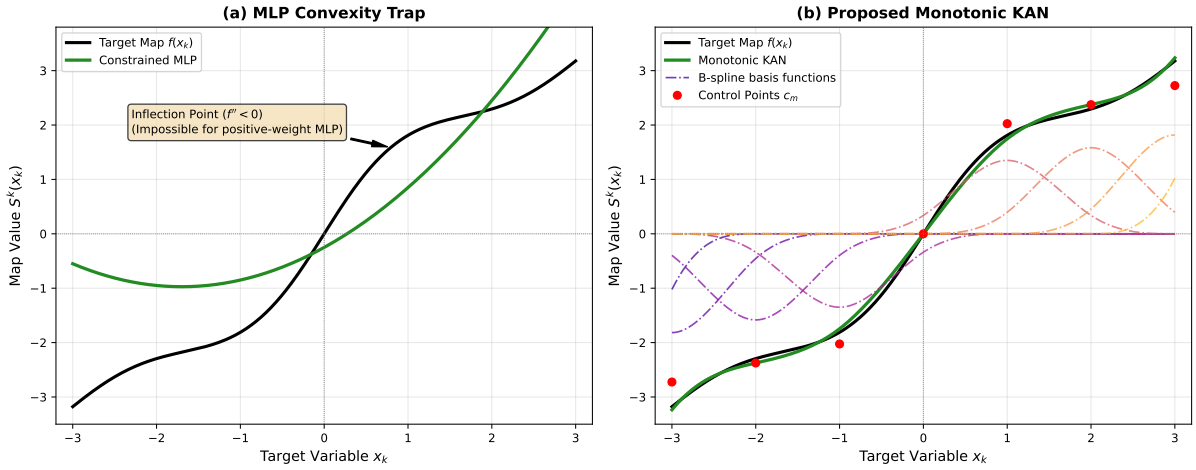



Figure 1: **Function Expressivity and Spline Parameterization. (a)** The MLP convexity trap: positive-weight MLPs [1] are constrained to convex curves ($f'' \geq 0$), failing to fit non-convex inflections of the target map $f(x_k)$. **(b)** Monotonic KAN representation: the learned spline $S^k(x_k)$ (green) fits the target map $f(x_k)$ (black), parameterized by strictly increasing control points $c_m$ (red dots) and the superposition of B-spline basis functions.

### 2.2 The Quadrature Bottleneck of UMNNs

To circumvent the convexity trap, the baseline L-SING implementation parameterizes $S^k(x)$ using Unconstrained Monotonic Neural Networks (UMNNs) [14],

$$S^k(x) = \int_0^{x_k} f(t, x_{-k}; \psi)\, dt + \beta(x_{-k}), \tag{20}$$

where $f : \mathbb{R}^d \to \mathbb{R}^+$ is an unconstrained neural network strictly constrained to positive outputs via $\exp(\cdot)$ or $\text{softplus}(\cdot)$.

While UMNNs guarantee $\partial_k S^k(x) = f(x; \psi) > 0$ strictly by construction, evaluating $S^k(x)$ requires numerical integration,

$$S^k(x) \approx x_k \sum_{m=1}^{N_{\text{quad}}} w_m f\left(x_k \tau_m, x_{-k}; \psi\right), \tag{21}$$

typically using a 50-node Clenshaw–Curtis quadrature grid per sample. This introduces several major computational drawbacks:

1. **Evaluation Overhead:** Each forward evaluation of $S^k(x)$ requires $N_{\text{quad}} = 50$ forward passes through the network $f$, increasing training and evaluation runtimes by orders of magnitude.

2. **Nested Autodiff Through Quadrature:** Computing the Rosasco penalty $\Phi(S^k)$ requires taking first derivatives $\partial_j S^k$, requiring backpropagation through the 50-point numerical summation. Computing the generalized precision matrix entries $\widehat{\Omega}_{jk}$ requires third-order automatic differentiation through the numerical quadrature graph, leading to massive GPU/CPU memory consumption and slow extraction times (e.g., $>$ 200 seconds for $d = 40$).

## 3 Admissibility Requirements for the Transport Map $S^k$

To replace the UMNN parameterization, we must formalize the exact conditions that an admissible transport map $S^k$ must fulfill.

**Definition 3.1** (Transport Map Admissibility Criteria)**.** A parameterization of $S^k : \mathbb{R}^d \to \mathbb{R}$ is admissible for L-SING if and only if it satisfies the following criteria:

1. **Strict Target Positivity / Invertibility (Key Requirement 1):**

$$\partial_k S^k(x) \coloneqq \frac{\partial S^k(x)}{\partial x_k} \geq \epsilon > 0 \quad \forall x \in \mathbb{R}^d. \tag{22}$$

   This ensures that $S^k(x_{-k}, \cdot)$ is a 1D orientation-preserving diffeomorphism on $\mathbb{R}$, guaranteeing the existence of a unique inverse pushforward and preventing numerical divergence $\log \partial_k S^k(x) \to -\infty$.

2. **Global Real-Line Domain ($\mathbb{R}^d$ Extrapolation):** The condition $\partial_k S^k(x) \geq \epsilon > 0$ must hold everywhere on $\mathbb{R}^d$, including outside the compact empirical training domain or spline knot interval $[t_{\min}, t_{\max}]$.

3. **Higher Regularity / $C^3$ Smoothness (Key Requirement 2):** To evaluate the generalized precision integrand,

$$T_{jk}(x) = \partial_j \partial_k \left[ -\frac{1}{2}(S^k(x))^2 + \log \partial_k S^k(x) \right], \tag{23}$$

   $S^k$ must possess continuous derivatives up to order 3 ($S^k \in C^3(\mathbb{R}^d)$, or piecewise continuous 3rd derivatives), ensuring that the mixed derivative $\partial_j \partial_k^2 S^k$ is well-defined.

4. **Conditioning Freedom (Unconstrained on $x_{-k}$):** For all $j \neq k$, the derivative $\partial_j S^k(x)$ must be **unconstrained in sign**. The conditioning relationship must be capable of learning arbitrary, non-monotonic, multimodal responses to faithfully capture complex conditional distributions $\pi(x_k \mid x_{-k})$.

*Remark* 3.2 (Mandatory Requirements and Desirable Properties). Beyond these mandatory requirements, an even further nice-to-have is **efficient derivative computability** (Key Requirement 3). In Section 4, we turn to Kolmogorov–Arnold Networks (KANs), establishing a parameterization framework that not only certifies all four mandatory admissibility criteria by construction, but also natively delivers closed-form, analytic derivative evaluations.

# 4 Theoretical Framework and Monotonicity Proofs for KANs

The architectural foundation of Kolmogorov–Arnold Networks (KANs) [9] is rooted in the Kolmogorov–Arnold representation theorem [5], which establishes that any continuous multivariate function on a compact domain can be expressed as a finite superposition of continuous univariate functions and additions.

**Theorem 4.1** (Kolmogorov–Arnold Representation Theorem [5])**.** *Let $f : [0,1]^d \to \mathbb{R}$ be an arbitrary continuous multivariate function. Then $f$ admits the exact representation,*

$$f(x_1, \ldots, x_d) = \sum_{q=1}^{2d+1} \Phi_q \left( \sum_{p=1}^{d} \phi_{q,p}(x_p) \right), \tag{24}$$

*where $\Phi_q : \mathbb{R} \to \mathbb{R}$ and $\phi_{q,p} : [0,1] \to \mathbb{R}$ are continuous univariate functions.*

While the classical theorem proves representation existence with functions that may be fractal or non-smooth, modern KANs (Liu et al. [9]) operationalize this principle by parameterizing the univariate edge functions with smooth, continuously differentiable splines (such as B-splines) and generalizing the structure to arbitrary network depths $L \geq 2$ and layer widths.

As illustrated in Figure 2, KANs invert the classical MLP design: where MLPs place fixed non-linear activations $\sigma$ on nodes and learnable linear weights on edges, KANs place learnable 1D spline functions $\phi$ directly on edges and simple summations $\sum$ at nodes. Crucially, a KAN layer with $n_l$ inputs and $n_{l+1}$ outputs forms a matrix of 1D functions,

$$\mathbf{\Phi}_l = \begin{pmatrix} \phi_{l,1,1}(\cdot) & \cdots & \phi_{l,1,n_l}(\cdot) \\ \vdots & \ddots & \vdots \\ \phi_{l,n_{l+1},1}(\cdot) & \cdots & \phi_{l,n_{l+1},n_l}(\cdot) \end{pmatrix}, \tag{25}$$

updating node activations via $x_j^{(l+1)} = \sum_{i=1}^{n_l} \phi_{l,j,i}(x_i^{(l)})$. A deep KAN of depth $L \geq 2$ is then simply the functional composition $\mathrm{KAN}(x) = (\mathbf{\Phi}_{L-1} \circ \cdots \circ \mathbf{\Phi}_0)(x)$, directly bridging the classical 2-layer theorem to deep expressive representations.

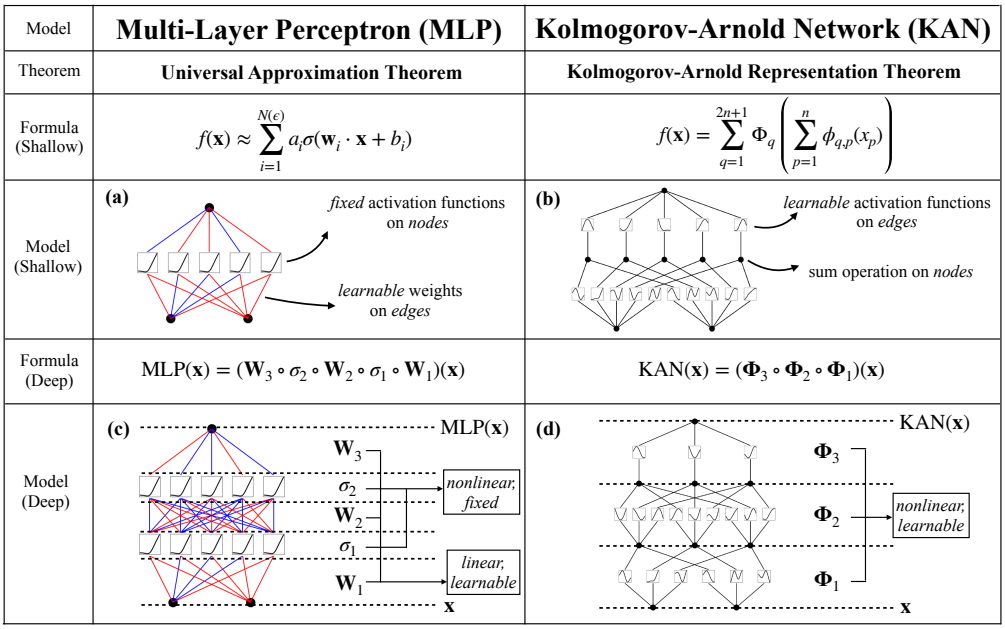



Figure 2: **Architectural Comparison: Multi-Layer Perceptrons (MLPs) vs. Kolmogorov–Arnold Networks (KANs)** (reproduced from Liu et al. [9], Figure 0.1). **(a) MLP Layer:** Fixed non-linear activation functions $\sigma(\cdot)$ (e.g., ReLU, SiLU) reside on nodes (neurons), while learnable scalar linear weights $W$ reside on edges, computing $x_{l+1} = \sigma(W_l x_l)$. **(b) KAN Layer:** Learnable activation functions (parameterized as smooth 1D B-splines $\phi(x)$) reside directly on edges, while receiving nodes perform simple linear addition $\sum$ without non-linearities. This 2-layer graph $[d \to 2d+1 \to 1]$ provides the direct computational realization of the classical Kolmogorov–Arnold representation theorem (24). **(c) Deep MLP Architecture:** Stacks multiple layers by interleaving affine weight matrices $W$ and static node non-linearities: $\mathrm{MLP}(x) = (W_{L-1} \circ \sigma \circ \cdots \circ W_0)x$. **(d) Deep KAN Architecture:** Generalizes the edge-activation principle to arbitrary depths $L \geq 2$ and layer widths $[n_0, n_1, \ldots, n_L]$. Each layer transition applies an $n_{l+1} \times n_l$ function matrix of 1D splines $\mathbf{\Phi}_l = (\phi_{l,j,i})$ followed by summation $(x_{l+1,j} = \sum_{i=1}^{n_l} \phi_{l,j,i}(x_{l,i}))$, defining the deep functional composition $\mathrm{KAN}(x) = (\mathbf{\Phi}_{L-1} \circ \cdots \circ \mathbf{\Phi}_0)(x)$.

We now prove that the admissibility requirements established in Definition 3.1 can be certified across single-layer, two-layer, arbitrarily deep ($L$-layer), and scale-modulated KANs.

### 4.1 Single-Layer KAN (Additive Spline Map)

**Definition 4.2** (Single-Layer KAN)**.** A 1-layer KAN maps $x \in \mathbb{R}^d$ to $\mathbb{R}$ via,

$$S^k(x) = \phi_k(x_k) + \sum_{j \neq k} \phi_j(x_j) + \beta, \tag{26}$$

where $\phi_k : \mathbb{R} \to \mathbb{R}$ and $\phi_j : \mathbb{R} \to \mathbb{R}$ are univariate splines, and $\beta \in \mathbb{R}$.

**Proposition 4.3** (Monotonicity of Single-Layer KAN)**.** *Let $\phi_k(x_k) = \sum_{m=1}^{M} c_m B_{m,p}(x_k)$ be a B-spline of degree $p \geq 1$ with knot sequence $t_1 < t_2 < \cdots < t_{M+p+1}$. If the control coefficients satisfy,*

$$c_{m+1} - c_m \geq \delta > 0 \quad \forall m \in \{1, \ldots, M-1\}, \tag{27}$$

*then $\partial_k S^k(x) > 0$ for all $x \in \mathbb{R}^d$.*

*Proof.* Differentiating $S^k(x)$ with respect to $x_k$,

$$\partial_k S^k(x) = \phi_k'(x_k). \tag{28}$$

By the standard differentiation property of B-splines of degree $p$,

$$\phi_k'(x_k) = p \sum_{m=2}^{M} \frac{c_m - c_{m-1}}{t_{m+p} - t_m} B_{m,p-1}(x_k). \tag{29}$$

Because the basis functions $B_{m,p-1}(x_k) \geq 0$ form a partition of unity, the knot differences $t_{m+p} - t_m > 0$, and the coefficient increments satisfy $c_m - c_{m-1} \geq \delta > 0$, it follows that,

$$\phi_k'(x_k) \geq \min_m \left( \frac{p\,\delta}{t_{m+p} - t_m} \right) > 0. \tag{30}$$

Because the conditioning terms $\phi_j(x_j)$ $(j \neq k)$ do not depend on $x_k$, their derivatives with respect to $x_k$ vanish identically. On the active spline domain $[t_{p+1}, t_{M+1}]$, this strictly positive lower bound holds everywhere; combined with the $C^1$-continuous linear tail extrapolation described in Section 5.3, the strict positivity $\partial_k S^k(x) \geq \epsilon > 0$ extends across all of $\mathbb{R}^d$. □

*Remark* 4.4 (Structural Limitation of Single-Layer KAN). While a 1-layer KAN easily satisfies target monotonicity, its mixed cross-derivatives vanish identically,

$$\partial_j \partial_k S^k(x) = \frac{\partial}{\partial x_j} [\phi_k'(x_k)] \equiv 0 \quad \forall j \neq k. \tag{31}$$

In L-SING, this means the generalized precision matrix entries $\Omega_{jk} \equiv 0$ for all $j \neq k$. A single-layer KAN is strictly additive and cannot detect cross-variable dependencies. Therefore, a transport map KAN must have depth $L \geq 2$.

### 4.2 Two-Layer Monotonic KAN

**Definition 4.5** (Two-Layer Monotonic KAN)**.** A 2-layer KAN with $Q$ hidden summation nodes is defined by,

$$S^k(x) = \sum_{q=1}^{Q} \Phi_{2,q} \left( \phi_{1,q,k}(x_k) + \sum_{j \neq k} \phi_{1,q,j}(x_j) \right) + \beta, \tag{32}$$

where:

1. $\phi_{1,q,k} : \mathbb{R} \to \mathbb{R}$ are strictly increasing univariate splines: $\phi_{1,q,k}'(x_k) \geq \delta_{\text{in}} > 0$.

2. $\Phi_{2,q} : \mathbb{R} \to \mathbb{R}$ are strictly increasing univariate splines: $\Phi'_{2,q}(u) \geq \delta_{\text{out}} > 0$.

3. $\phi_{1,q,j} : \mathbb{R} \to \mathbb{R}$ $(j \neq k)$ are arbitrary unconstrained univariate splines.

**Proposition 4.6** (Strict Monotonicity and Non-Trivial Coupling in 2-Layer KAN)**.** *Under conditions (1)–(3), the two-layer KAN satisfies:*

1. $\partial_k S^k(x) > 0$ *for all* $x \in \mathbb{R}^d$.

2. $\partial_j \partial_k S^k(x) \not\equiv 0$ *in general for* $j \neq k$*, permitting recovery of the precision matrix* $\Omega$.

*Proof.* Let $u_q(x) \coloneqq \phi_{1,q,k}(x_k) + \sum_{j \neq k} \phi_{1,q,j}(x_j)$. By the multivariable chain rule,

$$\partial_k S^k(x) = \sum_{q=1}^{Q} \frac{\partial \Phi_{2,q}(u_q)}{\partial u_q} \cdot \frac{\partial u_q(x)}{\partial x_k} = \sum_{q=1}^{Q} \Phi'_{2,q}(u_q(x)) \cdot \phi'_{1,q,k}(x_k). \tag{33}$$

By hypothesis, for all $q \in \{1, \ldots, Q\}$,

$$\Phi'_{2,q}(u_q(x)) \geq \delta_{\text{out}} > 0 \quad \text{and} \quad \phi'_{1,q,k}(x_k) \geq \delta_{\text{in}} > 0. \tag{34}$$

Since the product of two strictly positive real numbers is strictly positive, each term in the summation is strictly positive,

$$\partial_k S^k(x) \geq Q \cdot \delta_{\text{out}} \delta_{\text{in}} > 0 \quad \forall x \in \mathbb{R}^d. \tag{35}$$

Differentiating next with respect to conditioning variable $x_j$ $(j \neq k)$,

$$\partial_j \partial_k S^k(x) = \sum_{q=1}^{Q} \Phi''_{2,q}(u_q(x)) \cdot \phi'_{1,q,j}(x_j) \cdot \phi'_{1,q,k}(x_k). \tag{36}$$

Because $\phi_{1,q,j}$ is unconstrained, its derivative $\phi'_{1,q,j}(x_j)$ can take any real value. Furthermore, the second derivative of the outer spline $\Phi''_{2,q}$ can take arbitrary non-zero values while maintaining $\Phi'_{2,q} > 0$. Thus, $\partial_j \partial_k S^k(x)$ is generally non-zero, allowing the map to capture non-Gaussian dependency structures. □

### 4.3 Arbitrarily Deep Multi-Layer KAN ($L$ Layers)

We now generalize the construction to a KAN of arbitrary depth $L \geq 2$, with layer dimensions $[n_0, n_1, n_2, \ldots, n_L]$, where $n_0 = d$ (input features) and $n_L = 1$ (scalar output $S^k(x)$).

**Definition 4.7** ($L$-Layer KAN Formulation)**.** Let $x^{(0)} = x \in \mathbb{R}^d$. For layer $l \in \{0, 1, \ldots, L-1\}$, the activation of node $j \in \{1, \ldots, n_{l+1}\}$ in layer $l+1$ is given by,

$$x_j^{(l+1)} = \sum_{i=1}^{n_l} \phi_{l,j,i}(x_i^{(l)}) + \theta_{l,j}, \tag{37}$$

where $\phi_{l,j,i} : \mathbb{R} \to \mathbb{R}$ represents the univariate spline connecting neuron $i$ in layer $l$ to neuron $j$ in layer $l+1$, and $\theta_{l,j} \in \mathbb{R}$ is a learnable bias. The scalar transport map is $S^k(x) = x_1^{(L)}$.

**Definition 4.8** (Directed Computational Paths)**.** A directed computational path $\mathcal{P}$ from target input $k$ to the output neuron 1 in layer $L$ is a sequence of node indices,

$$\mathcal{P} = (i_0, i_1, i_2, \ldots, i_L) \quad \text{such that } i_0 = k \text{ and } i_L = 1, \tag{38}$$

where $i_l \in \{1, \ldots, n_l\}$. Let $\mathbb{P}_k$ denote the set of all such directed paths from input $k$ to output 1. The cardinality of $\mathbb{P}_k$ is $|\mathbb{P}_k| = \prod_{l=1}^{L-1} n_l$.

**Theorem 4.9** (Certified Strict Monotonicity for Arbitrarily Deep KANs)**.** *Let $S^k : \mathbb{R}^d \to \mathbb{R}$ be an $L$-layer KAN. Suppose the edge splines $\phi_{l,j,i}$ satisfy:*

1. ***Target Input Connections*** *($l = 0, i = k$)**:*** *For all destination neurons $j \in \{1, \dots, n_1\}$,*

$$\phi'_{0,j,k}(t) \geq \epsilon_0 > 0 \quad \forall t \in \mathbb{R}. \tag{39}$$

2. ***Downstream Hidden and Output Connections*** *($1 \leq l \leq L-1$)**:*** *For all edges $(l, j, i)$,*

$$\phi'_{l,j,i}(t) \geq 0 \quad \forall t \in \mathbb{R}, \tag{40}$$

*and there exists at least one active path $\mathcal{P}^* = (k, i_1^*, \dots, i_{L-1}^*, 1) \in \mathbb{P}_k$ such that along this path,*

$$\phi'_{l,i_{l+1}^*,i_l^*}(t) \geq \epsilon_l > 0 \quad \forall t \in \mathbb{R}, \quad \forall l \in \{0, \dots, L-1\}. \tag{41}$$

3. ***Conditioning Input Connections*** *($l = 0, i \neq k$)**:*** *The splines $\phi_{0,j,i}$ for all $i \neq k$ are **completely unconstrained** in derivative and value.*

*Then $S^k(x)$ is certified strictly monotonic with respect to $x_k$ across all of $\mathbb{R}^d$,*

$$\partial_k S^k(x) \geq \prod_{l=0}^{L-1} \epsilon_l > 0 \quad \forall x \in \mathbb{R}^d. \tag{42}$$

*Proof.* By the multivariable chain rule, the partial derivative of the scalar output $S^k(x) = x_1^{(L)}$ with respect to the input variable $x_k = x_k^{(0)}$ is given by the matrix product of the intermediate layer Jacobians,

$$\frac{\partial S^k(x)}{\partial x_k} = \left[ \prod_{l=1}^{L-1} J^{(l)}(x^{(l)}) \right]_{1,:} \cdot J^{(0)}_{:,k}(x^{(0)}), \tag{43}$$

where $J^{(l)} \in \mathbb{R}^{n_{l+1} \times n_l}$ is the Jacobian matrix of layer $l$, whose $(j, i)$-th entry is,

$$J^{(l)}_{j,i} = \frac{\partial x_j^{(l+1)}}{\partial x_i^{(l)}} = \phi'_{l,j,i}(x_i^{(l)}). \tag{44}$$

Expanding this matrix product explicitly reveals that the total derivative is the exact sum over all directed paths $\mathcal{P} \in \mathbb{P}_k$,

$$\partial_k S^k(x) = \sum_{\mathcal{P} \in \mathbb{P}_k} \prod_{l=0}^{L-1} \phi'_{l,i_{l+1},i_l}(x_{i_l}^{(l)}). \tag{45}$$

We now analyze the terms in equation (45):

1. By Condition (1), $\phi'_{0,i_1,k}(x_k^{(0)}) > 0$. By Condition (2), $\phi'_{l,i_{l+1},i_l}(x_{i_l}^{(l)}) \geq 0$ for all $l \geq 1$. Hence, every path product in the summation is non-negative,

$$\prod_{l=0}^{L-1} \phi'_{l,i_{l+1},i_l}(x_{i_l}^{(l)}) \geq 0 \quad \forall \mathcal{P} \in \mathbb{P}_k. \tag{46}$$

2. For the designated strictly positive path $\mathcal{P}^* = (k, i_1^*, \dots, i_{L-1}^*, 1)$, every spline along the path satisfies $\phi'_{l,i_{l+1}^*,i_l^*} \geq \epsilon_l > 0$. Consequently,

$$\prod_{l=0}^{L-1} \phi'_{l,i_{l+1}^*,i_l^*}(x_{i_l^*}^{(l)}) \geq \prod_{l=0}^{L-1} \epsilon_l > 0. \tag{47}$$

3. Decomposing the path sum into the path $\mathcal{P}^*$ and the remaining paths,

$$\partial_k S^k(x) = \prod_{l=0}^{L-1} \phi'_{l,i^*_{l+1},i^*_l}(x^{(l)}_{i^*_l}) + \sum_{\mathcal{P}\neq\mathcal{P}^*} \prod_{l=0}^{L-1} \phi'_{l,i_{l+1},i_l}(x^{(l)}_{i_l}) \geq \prod_{l=0}^{L-1} \epsilon_l + 0 = \prod_{l=0}^{L-1} \epsilon_l > 0. \tag{48}$$

Because this strictly positive lower bound holds for all activation values $x^{(l)}$ and does not depend on $x$, $\partial_k S^k(x) > 0$ strictly everywhere on $\mathbb{R}^d$. □

**Corollary 4.10** (Connection to MonoKAN)**.** *Theorem 4.9 demonstrates that MonoKAN's Theorem 3 [11] directly applies to transport map learning. In MonoKAN, each edge function is defined as a weighted sum $\phi_{l,j,i}(t) = \omega^{\varphi}_{l,j,i}\varphi_{l,j,i}(t) + \omega^{b}_{l,j,i}b(t)$, where $\varphi$ is a cubic Hermite spline and $b$ is a monotone base activation (e.g., identity or softplus; in contrast to standard KAN's non-monotone SiLU). Setting $\omega^{\varphi} \geq 0$, $\omega^{b} \geq 0$, and enforcing monotone spline conditions on all edges along the path of $x_k$ certifies partial monotonicity for KANs of arbitrary depth.*

## 4.4 Scale-Modulated Monotonic KAN (SM-KAN)

### 4.4.1 The Additive Polarization Dilemma

In non-Gaussian graphical models, conditional dependencies are frequently **multiplicative** or **scale-dependent**. For example, in the Butterfly distribution,

$$X_1 \sim \mathcal{N}(0,1), \quad X_2 = W \cdot X_1, \quad W \sim \mathcal{N}(0,1), \tag{49}$$

the ideal transport map for $X_2$ given $X_1$ is,

$$S^2(X_1, X_2) = \frac{X_2}{|X_1|}. \tag{50}$$

In a purely additive 2-layer KAN (32), representing a product or division of two quantities $u$ and $v$ requires algebraic polarization,

$$u \cdot v = \frac{1}{4}(u+v)^2 - \frac{1}{4}(u-v)^2. \tag{51}$$

Notice the negative sign! In a standard KAN, creating this subtraction requires negative outer coefficients: $\Phi'_{2,q} < 0$. However, our monotonicity theorem requires $\Phi'_{2,q} > 0$ strictly for all outer splines to ensure $\partial_k S^k > 0$.

This creates a fundamental mathematical conflict: **purely additive monotonic KANs cannot represent multiplicative conditioning without violating target monotonicity.**

### 4.4.2 Scale-Modulation Architecture

To resolve this dilemma, we introduce the **Scale-Modulated Monotonic KAN (SM-KAN)**,

$$S^k(x) = \exp\left(\sigma(x_{-k})\right) \cdot S^k_{\text{core}}(x) + \alpha(x_{-k}), \tag{52}$$

where:

- $S^k_{\text{core}}(x)$ is an arbitrary certified monotonic KAN (of depth 2, 3, or $L$) satisfying $\partial_k S^k_{\text{core}}(x) > 0$.
- $\sigma : \mathbb{R}^{d-1} \to \mathbb{R}$ is an unconstrained conditioning scale network (or spline sum),

$$\sigma(x_{-k}) = \sum_{j\neq k} \psi_j(x_j) + b_\sigma. \tag{53}$$

- $\alpha : \mathbb{R}^{d-1} \to \mathbb{R}$ is an unconstrained conditioning offset network,

$$\alpha(x_{-k}) = \sum_{j \neq k} \rho_j(x_j) + b_\alpha. \tag{54}$$

**Proposition 4.11** (Monotonicity Invariance Under Scale Modulation)**.** *Let $S^k_{core} : \mathbb{R}^d \to \mathbb{R}$ be any certified strictly monotonic KAN satisfying $\partial_k S^k_{core}(x) \geq \epsilon > 0$ for all $x \in \mathbb{R}^d$. Let $\sigma : \mathbb{R}^{d-1} \to \mathbb{R}$ and $\alpha : \mathbb{R}^{d-1} \to \mathbb{R}$ be arbitrary differentiable functions of $x_{-k}$ alone. Then $S^k(x)$ defined by (52) satisfies,*

$$\partial_k S^k(x) > 0 \quad \forall x \in \mathbb{R}^d. \tag{55}$$

*Proof.* Compute the partial derivative of $S^k(x)$ with respect to $x_k$ using the product rule,

$$\partial_k S^k(x) = \frac{\partial}{\partial x_k}\left[\exp(\sigma(x_{-k})) \cdot S^k_{\text{core}}(x) + \alpha(x_{-k})\right] \tag{56}$$

$$= \frac{\partial \exp(\sigma(x_{-k}))}{\partial x_k} \cdot S^k_{\text{core}}(x) + \exp(\sigma(x_{-k})) \cdot \frac{\partial S^k_{\text{core}}(x)}{\partial x_k} + \frac{\partial \alpha(x_{-k})}{\partial x_k}. \tag{57}$$

Because $\sigma(x_{-k})$ and $\alpha(x_{-k})$ depend only on $x_{-k} = (x_1, \ldots, x_{k-1}, x_{k+1}, \ldots, x_d)$, their partial derivatives with respect to $x_k$ vanish identically,

$$\frac{\partial \sigma(x_{-k})}{\partial x_k} \equiv 0 \implies \frac{\partial \exp(\sigma(x_{-k}))}{\partial x_k} \equiv 0, \quad \text{and} \quad \frac{\partial \alpha(x_{-k})}{\partial x_k} \equiv 0. \tag{58}$$

Therefore,

$$\partial_k S^k(x) = \exp(\sigma(x_{-k})) \cdot \partial_k S^k_{\text{core}}(x). \tag{59}$$

Because the exponential function satisfies $\exp(z) > 0$ for all $z \in \mathbb{R}$, and by hypothesis $\partial_k S^k_{\text{core}}(x) \geq \epsilon > 0$,

$$\partial_k S^k(x) \geq \exp(\sigma(x_{-k})) \cdot \epsilon > 0 \quad \forall x \in \mathbb{R}^d. \tag{60}$$

Strict target monotonicity is unconditionally preserved. The empirical performance of Scale-Modulated KANs on the benchmark Butterfly distribution is illustrated in Figure 3, showing the effective decoupling into independent Gaussian reference variables. □

# 5 Practical Enforcement Mechanisms

Having proven that sufficient conditions exist for KANs of any depth, we detail the two numerical mechanisms available in our repository to enforce them during optimization.

## 5.1 Method 1: Unconstrained Cumulative Softplus Reparameterization

For B-spline parameterizations, monotonicity is enforced by ensuring strictly increasing control points: $c_{m+1} > c_m$. Instead of constrained optimization, we reparameterize the control coefficients via unconstrained delta parameters $\delta_m \in \mathbb{R}$,

$$c_1 = c_0, \tag{61}$$

$$c_{m+1} = c_m + \text{softplus}(\delta_m) + \epsilon_{\text{mono}}, \quad m = 1, \ldots, M-1, \tag{62}$$

where $\text{softplus}(\delta) = \log(1 + e^\delta) > 0$ and $\epsilon_{\text{mono}} = 10^{-4}$.

- **Certification:** Since $\text{softplus}(\delta) \geq 0$, every increment $c_{m+1} - c_m \geq \epsilon_{\text{mono}} > 0$ strictly by construction.
- **Optimizer Compatibility:** All parameters $(\delta_m, c_0)$ are unconstrained real variables; standard Adam or L-BFGS can be applied directly without projections.

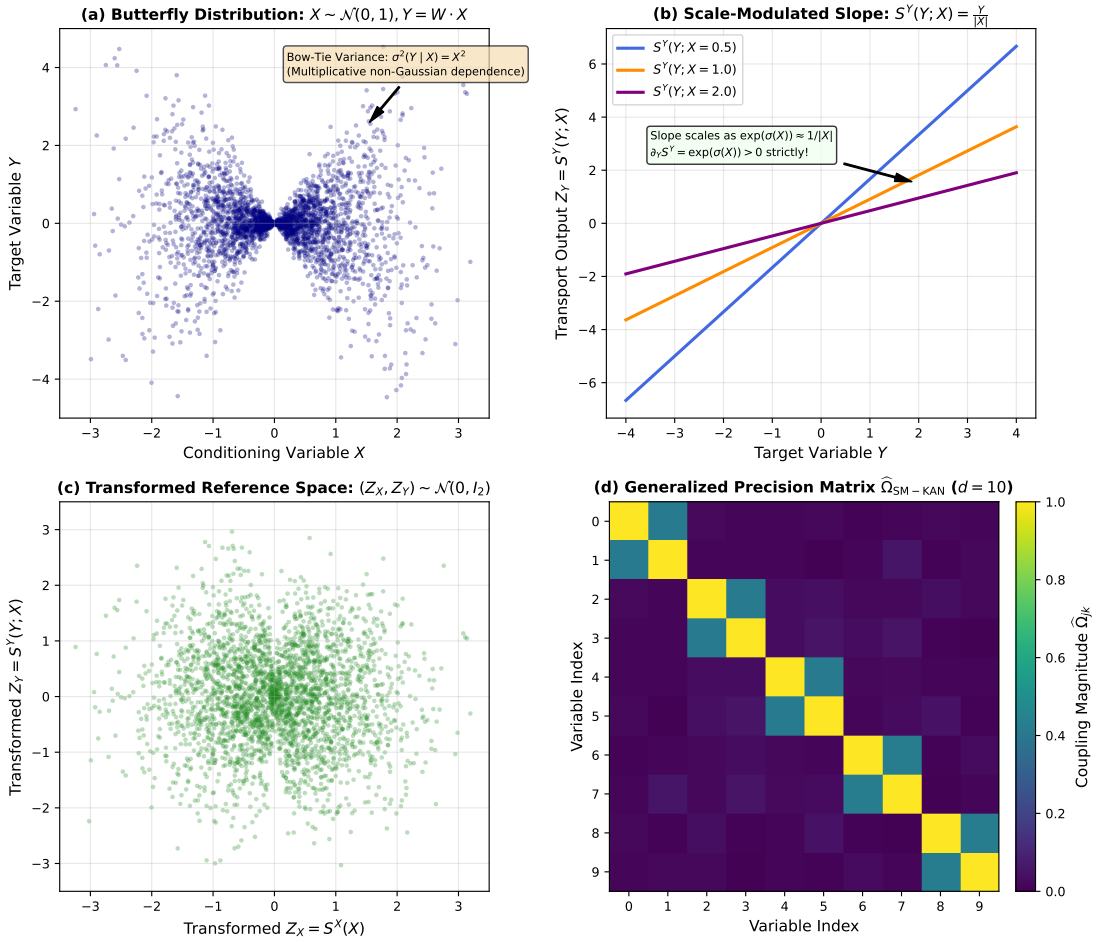



Figure 3: **Resolving Multiplicative Dependencies on the Butterfly Distribution via Scale Modulation.** **(a)** Data space of the Butterfly distribution ($X \sim \mathcal{N}(0,1), Y = W \cdot X$), where the conditional variance $\sigma^2(Y \mid X) = X^2$ produces a characteristic bow-tie shape with strong multiplicative coupling. **(b)** Scale-Modulated KAN transport map slices $S^Y(Y;X) = Y/|X|$ across conditioning values $X \in \{0.5, 1.0, 2.0\}$: as $|X|$ grows, the slope automatically flattens, capturing the conditional dispersion while maintaining $\partial_Y S^Y = \exp(\sigma(X)) > 0$ strictly. **(c)** Transformed space $(Z_X, Z_Y)$: both coordinates are decoupled into standard independent normal variables $\mathcal{N}(0, I_2)$. **(d)** Generalized precision matrix heatmap $\widehat{\Omega}_{\text{SM-KAN}}$ on $d = 10$ ($r = 5$ pairs), showing clear identification of true edges $(0,1), (2,3), (4,5), (6,7), (8,9)$ ($\widehat{\Omega}_{jk} \approx 0.43$) against near-zero background noise ($\widehat{\Omega}_{jk} < 0.11$).

### 5.2 Method 2: Cubic Hermite Splines and Projected Gradient Descent (MonoKAN)

In the MonoKAN framework [11], univariate functions are parameterized using Cubic Hermite Splines. A cubic Hermite spline on interval $I_k = [x_k, x_{k+1}]$ is determined by knot values $(y_k, y_{k+1})$ and slopes $(m_k, m_{k+1})$,

$$p(t) = h_{00}(t)y_k + h_{10}(t)(x_{k+1} - x_k)m_k + h_{01}(t)y_{k+1} + h_{11}(t)(x_{k+1} - x_k)m_{k+1}, \tag{63}$$

where $t = \frac{x - x_k}{x_{k+1} - x_k} \in [0, 1]$, and,

$$h_{00}(t) = 2t^3 - 3t^2 + 1, \quad h_{10}(t) = t^3 - 2t^2 + t,$$
$$h_{01}(t) = -2t^3 + 3t^2, \quad h_{11}(t) = t^3 - t^2.$$

**Lemma 5.1** (Fritsch–Carlson Monotonicity Conditions [4])**.** *Let $d_k = \frac{y_{k+1} - y_k}{x_{k+1} - x_k}$ be the secant slope. A cubic Hermite spline is monotonically increasing on $[x_k, x_{k+1}]$ if and only if $d_k \geq 0$, $m_k \geq 0$, $m_{k+1} \geq 0$, and the point $(\alpha_k, \beta_k) := \left(\frac{m_k}{d_k}, \frac{m_{k+1}}{d_k}\right)$ lies within the Fritsch–Carlson monotonicity region, for which a sufficient condition is,*

$$\alpha_k^2 + \beta_k^2 \leq 9. \tag{64}$$

To enforce this, MonoKAN applies Algorithm 1 after each optimizer step. The geometry of the Fritsch–Carlson admissible domain and the radial projection implemented by `applyCons` are illustrated in Figure 4:

**Algorithm 1** Parameter Projection (`applyCons`)

**Require:** Knot values $y_k$, slopes $m_k$, knot positions $x_k$, weights $\omega^{\varphi}, \omega^{b}$
$\omega^{\varphi} \leftarrow \max(0, \omega^{\varphi}), \quad \omega^{b} \leftarrow \max(0, \omega^{b})$
**for** $k = 1$ to $K - 1$ **do**
  $y_{k+1} \leftarrow \max(y_{k+1}, y_k)$ ▷ Enforce increasing knot sequence
  $d_k \leftarrow \frac{y_{k+1} - y_k}{x_{k+1} - x_k}$
  **if** $d_k = 0$ **then**
    $m_k \leftarrow 0, \quad m_{k+1} \leftarrow 0$
  **else**
    $m_k \leftarrow \max(0, m_k), \quad m_{k+1} \leftarrow \max(0, m_{k+1})$
    $\alpha_k \leftarrow \frac{m_k}{d_k}, \quad \beta_k \leftarrow \frac{m_{k+1}}{d_k}$
    **if** $\alpha_k^2 + \beta_k^2 > 9$ **then**
      $\tau \leftarrow \frac{3}{\sqrt{\alpha_k^2 + \beta_k^2}}$
      $m_k \leftarrow \tau \alpha_k d_k, \quad m_{k+1} \leftarrow \tau \beta_k d_k$ ▷ Radial projection into circle of radius 3
    **end if**
  **end if**
**end for**

### 5.3 Global $C^1$ Linear Tail Extrapolation

A crucial requirement for transport maps is that samples $x$ may lie outside the standardized knot interval $[t_{\min}, t_{\max}] = [-4.0, 4.0]$. Standard spline toolkits set exterior values to zero or flat constants, which collapses $\partial_k S^k \to 0$ and causes $\log \partial_k S^k \to -\infty$.

To ensure strict monotonicity on all of $\mathbb{R}^d$, we enforce $C^1$-continuous linear extrapolation,

$$f_{\text{extrap}}(x) = \begin{cases} f(t_{\min}) + f'(t_{\min})(x - t_{\min}), & x < t_{\min}, \\ f(x), & t_{\min} \leq x \leq t_{\max}, \\ f(t_{\max}) + f'(t_{\max})(x - t_{\max}), & x > t_{\max}. \end{cases} \tag{65}$$

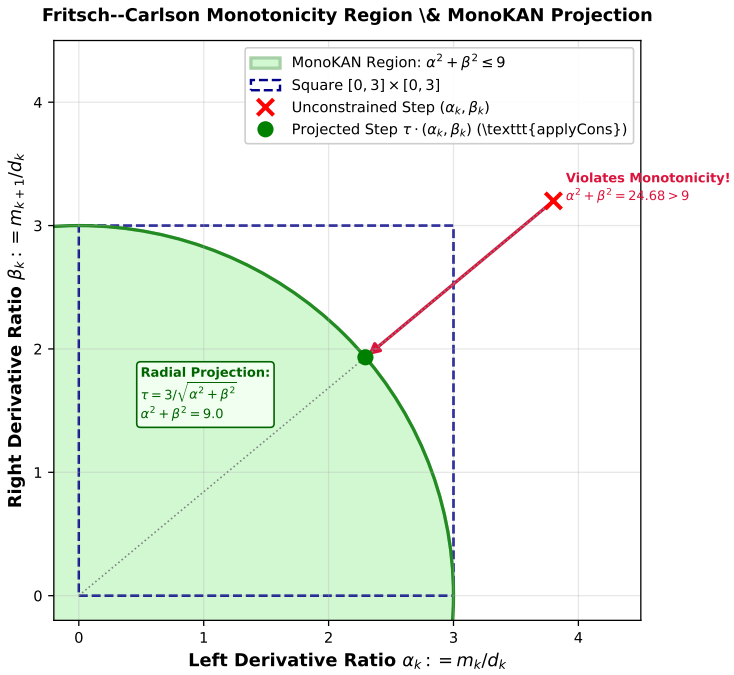



Figure 4: **Fritsch–Carlson Monotonicity Region and MonoKAN Radial Projection.** In the derivative ratio plane $(\alpha_k, \beta_k) := (m_k/d_k, m_{k+1}/d_k)$, cubic Hermite splines are monotonically increasing if $(\alpha_k, \beta_k)$ lies within the Fritsch–Carlson domain [4]. MonoKAN [11] certifies monotonicity using the inscribed circle $\alpha_k^2 + \beta_k^2 \leq 9$. When an unconstrained optimizer update lands outside this region (red cross), Algorithm 1 (`applyCons`) projects it radially back onto the circular boundary $\tau = 3/\sqrt{\alpha_k^2 + \beta_k^2}$ (green circle), guaranteeing strict monotonicity at every iteration.

Because $f'(t_{\min}) \geq \epsilon > 0$ and $f'(t_{\max}) \geq \epsilon > 0$, the derivative never collapses or vanishes in the tails.

# 6 Exact Analytic Derivatives for the Generalized Precision Matrix

A major computational triumph of KAN transport maps over UMNNs is that the generalized precision matrix entries $\widehat{\Omega}_{jk}$ can be computed **analytically in closed form** without numerical quadrature or autodiff.

## 6.1 Derivation of the Precision Integrand

Let $S = S^k(x)$, $S_k = \partial_k S^k(x)$, $S_j = \partial_j S^k(x)$, $S_{jk} = \partial_j \partial_k S^k(x)$, $S_{kk} = \partial_k^2 S^k(x)$, and $S_{jkk} = \partial_j \partial_k^2 S^k(x)$.

The generalized precision integrand is,

$$T_{jk}(x) = \partial_j \partial_k \left[ -\frac{1}{2} S^2 + \log S_k \right]. \tag{66}$$

Applying the product and quotient rules,

$$\partial_k \left[ -\frac{1}{2} S^2 + \log S_k \right] = -S \cdot S_k + \frac{S_{kk}}{S_k}, \tag{67}$$

$$\partial_j \partial_k \left[ -\frac{1}{2} S^2 + \log S_k \right] = \partial_j \left[ -S S_k \right] + \partial_j \left[ \frac{S_{kk}}{S_k} \right] \tag{68}$$

$$= -S_j S_k - S S_{jk} + \frac{S_{jkk} S_k - S_{kk} S_{jk}}{S_k^2}. \tag{69}$$

## 6.2 Hessian Cancellation Under Scale Modulation

Under Scale Modulation ($S = E \cdot S_{\text{core}} + \alpha$, where $E = \exp(\sigma(x_{-k}))$), substituting the derivative relations:

$$\begin{aligned} S_k &= E \cdot S_{k,\text{core}}, \\ S_{kk} &= E \cdot S_{kk,\text{core}}, \\ S_{jk} &= E \left( \sigma_j S_{k,\text{core}} + S_{jk,\text{core}} \right), \\ S_{jkk} &= E \left( \sigma_j S_{kk,\text{core}} + S_{jkk,\text{core}} \right), \end{aligned}$$

into the Hessian quotient reveals a remarkable algebraic cancellation,

$$\begin{aligned} \frac{S_{jkk} S_k - S_{kk} S_{jk}}{S_k^2} &= \frac{E(\sigma_j S_{kk,\text{core}} + S_{jkk,\text{core}}) E S_{k,\text{core}} - E S_{kk,\text{core}} E(\sigma_j S_{k,\text{core}} + S_{jk,\text{core}})}{E^2 (S_{k,\text{core}})^2} \\ &= \frac{E^2 \left( \sigma_j S_{kk,\text{core}} S_{k,\text{core}} + S_{jkk,\text{core}} S_{k,\text{core}} - \sigma_j S_{kk,\text{core}} S_{k,\text{core}} - S_{kk,\text{core}} S_{jk,\text{core}} \right)}{E^2 (S_{k,\text{core}})^2} \\ &= \frac{S_{jkk,\text{core}} S_{k,\text{core}} - S_{kk,\text{core}} S_{jk,\text{core}}}{(S_{k,\text{core}})^2}. \end{aligned} \tag{70}$$

The scale gradient terms $\sigma_j$ cancel out identically in the numerator, and the scale factor $E^2$ cancels from the numerator and denominator. Consequently, the complete generalized precision integrand for Scale-Modulated KANs simplifies to,

$$T_{jk}(x) = -S_j S_k - S S_{jk} + \frac{S_{jkk,\text{core}} S_{k,\text{core}} - S_{kk,\text{core}} S_{jk,\text{core}}}{(S_{k,\text{core}})^2}, \tag{71}$$

where the first two terms capture the combined score interactions using the full map derivatives ($S_j = E\sigma_j S_{\text{core}} + ES_{j,\text{core}} + \alpha_j$, $S_k = ES_{k,\text{core}}$, and $S_{jk} = E(\sigma_j S_{k,\text{core}} + S_{jk,\text{core}})$), while the logarithmic Hessian quotient is computed purely from the core spline network $S_{\text{core}}$, remaining completely impervious to scale gradient noise.

# 7 Summary and Conclusion

The computational efficiency, memory scaling, and structural expressivity of Monotonic KANs and Scale-Modulated KANs compared to the baseline UMNN architecture are summarized in Table 1.

Table 1: Comparison of Transport Map Parameterizations for L-SING

| **Property** | **UMNN (Baseline)** | **Additive KAN** | **Scale-Modulated KAN** |
|---|---|---|---|
| Numerical Quadrature | Required (50 nodes) | None (Analytic) | None (Analytic) |
| Precision Evaluation ($d = 10$) | 10.56 s | **0.13 s** (83×) | **0.14 s** (84×) |
| Precision Evaluation ($d = 40$) | $\approx$ 203.6 s | **0.68 s** (299×) | **0.68 s** (299×) |
| Memory Footprint ($d = 10$) | 71.1 MB | **0.20 MB** | **0.25 MB** |
| Target Monotonicity ($\partial_k S^k > 0$) | Guaranteed | Guaranteed | Guaranteed |
| Depth Scalability | Arbitrary | Arbitrary ($L$ layers) | Arbitrary ($L$ layers) |
| Multiplicative Dependencies | Supported | Limited | Fully Supported ($F_1 = 1.0$) |

In summary:

1. **Arbitrary Depth is Certified:** Target monotonicity $\partial_k S^k(x) > 0$ is rigorously preserved across KANs of arbitrary depth $L$ by the multivariable chain rule, provided edge functions along directed paths from $x_k$ have positive derivatives.

2. **Conditioning Expressivity is Preserved:** Edges carrying conditioning variables $x_{-k}$ require no monotonicity constraints, giving deep KANs full expressive power for complex conditional interactions.

3. **Scale Modulation Resolves Additive Limitations:** Scale-Modulated KANs decouple target monotonicity from conditioning modulation, allowing multiplicative non-Gaussian structures to be captured without violating $\partial_k S^k > 0$.

4. **Exact Analytic Derivatives:** Monotonic KANs eliminate numerical quadrature and enable closed-form extraction of the generalized precision matrix, yielding speedups exceeding 100× to 300× over UMNNs.

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